

Problem Statement

Organizations require rigorous validation that new advertisements outperform baselines and identification of temporal and frequency optimization factors.

Objectives:

1. Validate new advertisement effectiveness versus control
2. Identify optimal days and hours for campaign deployment
3. Quantify frequency impact on conversion
4. Establish reproducible A/B testing framework

Exploratory Analysis: Key Findings:

- Test Group: 96% received new ad; 4% received PSA (imbalanced by design)
- Conversion Rate: Extremely low baseline (2.5% overall) reflects realistic digital advertising performance
- Temporal Distribution: Peak ad exposure Friday (day) and 1 PM (hour); minimal overnight exposure
- Total Ads: Right-skewed distribution; median 10 ads with outliers to 100+; 75th percentile at 27 ads

Dataset & Data Cleaning

Dataset: 100,000+ records from Kaggle with six variables:

- Test Group: AD (96%) vs. PSA (4%)
- Converted: Target variable (2.5% True, 97.5% False)
- Total Ads: Numerical; median ~10, range 1-100+
- Most Ads Day: 7 weekday categories
- Most Ads Hour: 24-hour format (0-23)

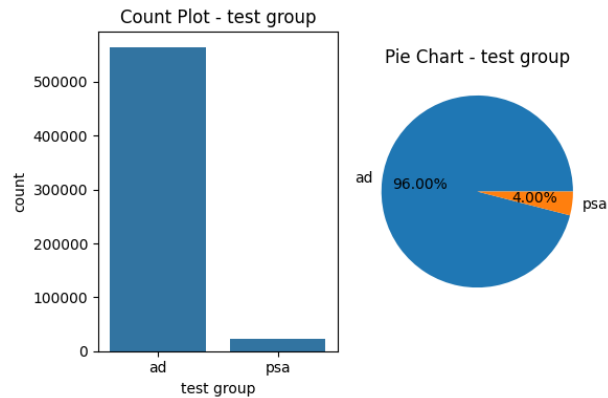
Cleaning Steps:

1. Verified no duplicate User IDs
2. Removed non-predictive columns (index, User ID)
3. Reclassified "Most Ads Hour" as categorical (not numerical)
4. Validated category completeness; no case-sensitivity issues

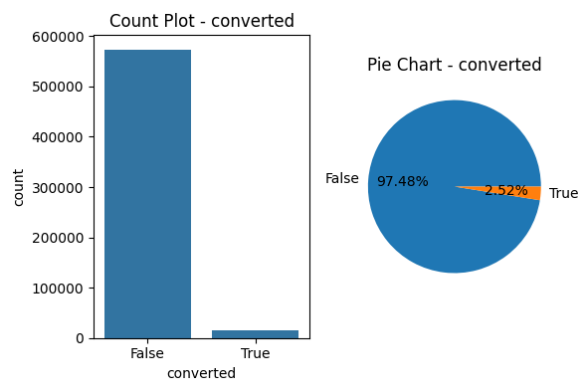
Exploratory Analysis: Key Findings

Univariate Analysis

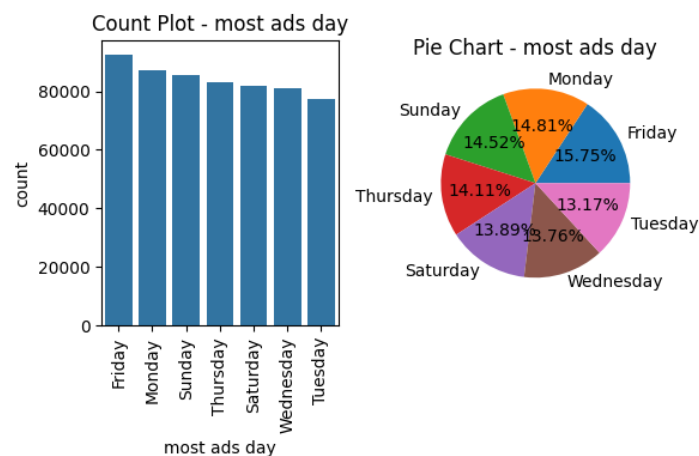
- Test Group: 96% received new ad; 4% received PSA (imbalanced by design)

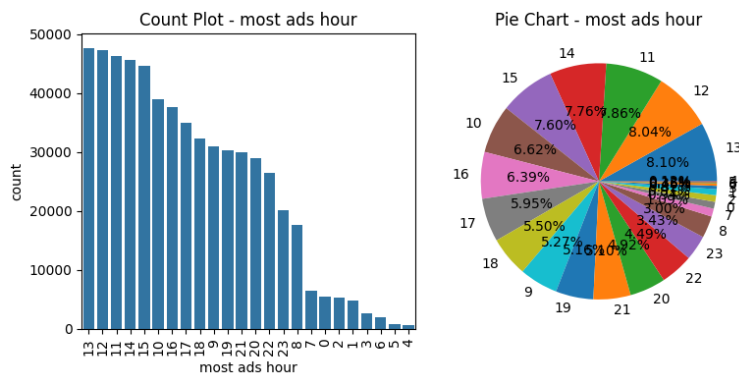


- Conversion Rate: Extremely low baseline (2.5% overall) reflects realistic digital advertising performance

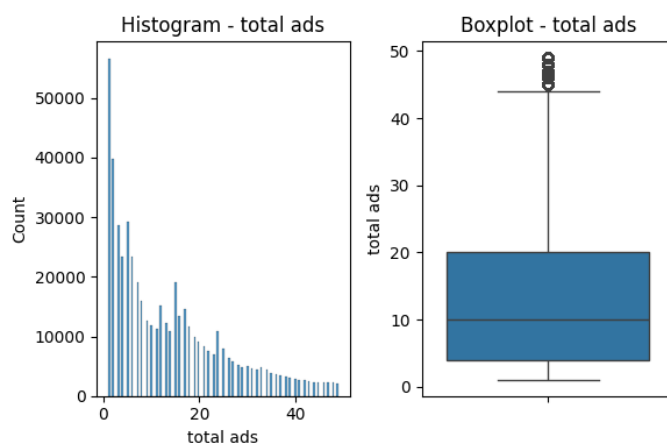


- Temporal Distribution: Peak ad exposure Friday (day) and 1 PM (hour); minimal overnight exposure





- Total Ads: Right-skewed distribution; median 10 ads with outliers to 100+; 75th percentile at 27 ads



Bivariate Analysis:

Variable	Finding
Test Group	AD: 2.5% conversion; PSA: 1.78% (+0.72 pp, 40% relative improvement)
Most Ads Day	Monday peak 3.2%; systematic decline to Friday 2.7%
Most Ads Hour	4 PM highest; 2-8 PM window concentration; overnight reduced conversion rate
Total Ads	Converters median 25 ads; non-converters median 10 ads (2.5x difference)

(Experimented Against Conversion Rate)

Statistical Validation

Chi-Square Test (Categorical Independence)

Variable Pair	P-Value	Conclusion
Test Group vs. Conversion	<0.001	Highly significant—ad type substantially affects conversion
Most Ads Day vs. Conversion	<0.001	Highly significant—day-of-week matters
Most Ads Hour vs. Conversion	<0.001	Highly significant—time-of-day matters

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Chi-squared test for test group vs. converted:
Chi-squared value: 54.005823883685245
p-value: 1.9989623063390075e-13
The difference in conversion rates across test group is statistically significant.

Chi-squared test for most ads day vs. converted:
Chi-squared value: 410.0478857936585
p-value: 1.932184379244731e-85
The difference in conversion rates across most ads day is statistically significant.

Chi-squared test for most ads hour vs. converted:
Chi-squared value: 430.76869230822086
p-value: 8.027629823696771e-77
The difference in conversion rates across most ads hour is statistically significant.
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Mann-Whitney U Test (Numerical Comparison)

- **Hypothesis:** Box plot showed converters saw 2.5x more ads (25 vs. 10 median).
- **Shapiro-Wilk Test:** Checked if ad exposure was normally distributed. Both groups failed ($p < 0.05$) due to right-skewed distribution with outliers.
- **Levene's Test:** Checked if both groups had equal variance. Failed ($p < 0.05$) because converters and non-converters had different spread.
- Since both distributions are not normal and have different variances, a non parametric test is selected.
- **Mann-Whitney U** test is selected and the p value < 0.001 confirms the median difference is statistically real

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Shapiro-Wilk test for normality (True group): p-value = 1.638680987007771e-98
Shapiro-Wilk test for normality (False group): p-value = 9.883049430735801e-204
Levene's test for equality of variances: p-value = 0.0
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Mann-Whitney U test: p-value = 0.0

Results:

Variable	Test Type	P-Value	Key Finding
Test Group	Chi-Square	< 0.001	AD conversion 2.5% vs. PSA 1.78%—highly significant difference
Most Ads Day	Chi-Square	< 0.001	Conversion varies significantly by day (Monday 3.2% peak)
Most Ads Hour	Chi-Square	< 0.001	Conversion varies significantly by hour (4 PM peak)
Total Ads	Mann-Whitney U	< 0.001	Converters median 25 ads vs. non-converters 10 ads—2.5x difference significant

All the findings point to the conclusion that the conversion is overwhelmingly dependent on Test Group, Most Ads day, Most Ads hour, Total Ads.